Notes: Manso (JFE, 2013)

Feedback Effects of Credit Ratings

Contents

1 Big picture

- **Question.** How should a credit rating agency assign *accurate* ratings when ratings themselves change the borrower's funding cost and therefore change the borrower's probability of default?
- **Setting.** The paper builds a continuous-time model in which a firm's cash flow follows a diffusion process, the firm has ratings-based performance-sensitive debt, equityholders choose when to default, and the rating agency chooses ratings so that they match target default probabilities over a fixed horizon.
- **Core challenge.** There is a circularity problem. A downgrade raises coupon payments, higher coupon payments make default more attractive to equityholders, earlier default raises true default probabilities, and that in turn justifies lower ratings. The rating agency is therefore not just measuring credit quality; it is helping determine it.
- **Main conceptual contribution.** The paper turns the interaction between the firm and the rating agency into a game of strategic complementarity. Even if the agency only wants accuracy and has no exogenous bias, the model can generate multiple self-consistent accurate rating policies.
- **Main model design.** The firm chooses a default threshold in cash-flow space. The rating agency chooses rating thresholds in the same cash-flow space. Ratings are "accurate" when they correctly classify the firm's default probability over horizon T relative to exogenous benchmark thresholds $G_0 > G_1 > \cdots > G_I$.
- **Main equilibrium result.** There is always a smallest Markov equilibrium and a largest Markov equilibrium. The **soft-rating-agency equilibrium** has higher ratings, lower coupon payments, lower default probability, and lower bankruptcy costs. The **tough-rating-agency equilibrium** has the opposite properties.
- **Main welfare result.** Because ratings are assumed accurate in equilibrium, the only welfare loss in the model comes from bankruptcy costs. Therefore, the soft equilibrium Pareto-dominates the tough equilibrium.
- **Key policy result.** A small issuer-paid fee can improve equilibrium selection by giving the rating agency a second-order concern for borrower survival, which pushes it toward the soft equilibrium among all accurate equilibria. By contrast, stress tests tied to rating triggers can push the system toward the tough equilibrium.
- **Key dynamics result.** When equilibrium is unique, best-response dynamics are globally stable and ratings move gradually. When multiple equilibria exist, small shocks can trigger multi-notch downgrades or even immediate default — the paper's "credit-cliff dynamic."
- **Key competition result.** With two agencies, competition does *not* automatically improve outcomes. If debt contracts use the *minimum* rating when agencies disagree, the unique equilibrium is the tough equilibrium. If they use the *maximum* rating, the unique equilibrium is the soft equilibrium.
- **Economic takeaway.** Tougher ratings, more aggressive stress tests, or more competition need not improve welfare once ratings feed back into fundamentals. In this setting, softer ratings can

be just as accurate and strictly better for welfare.

2 Model environment and key objects

2.1 Cash-flow process and firm value

The firm generates nonnegative cash flow at rate δ_t. Fundamentals evolve according to the diffusion

$$d\delta_t = \mu(\delta_t)\,dt + \sigma(\delta_t)\,dB_t, \tag{1}$$

where B_t is standard Brownian motion.

Agents are risk neutral and discount at the risk-free rate r. The value of the firm's assets is

$$A_t = E_t\left[\int_t^{\infty} e^{-r(s-t)}\delta_s\,ds\right]. \tag{2}$$

The paper assumes conditions on $\mu(\cdot)$ and $\sigma(\cdot)$ that guarantee a unique strong solution and finite firm value.

Interpretation. The only state variable is current cash flow δ_t. Better current fundamentals raise continuation value and lower short-horizon default risk.

2.2 Ratings-based performance-sensitive debt

Credit ratings take values in $\{1,\dots,I\}$, where 1 is the lowest rating and I is the highest. The firm has ratings-based performance-sensitive debt (PSD) with coupon schedule $C(\cdot)$ satisfying

$$C(i) \geq C(i+1). \tag{3}$$

So a lower rating implies a higher promised payment rate.

The paper interprets this broadly:

- **Explicit PSD:** rating triggers in bank loans or other contracts directly increase promised coupons after a downgrade.
- **Implicit PSD:** the rollover rate on short-term debt becomes higher when ratings deteriorate.

Interpretation. This debt structure is the entire source of the feedback effect. If ratings did not affect financing costs, the agency would only be measuring default risk rather than helping create it.

2.3 The firm's default problem and equity value

Given a rating process R, equityholders choose a default time $\hat{\tau}$ to maximize initial equity value:

$$W_0 = \sup_{\hat{\tau} \in \mathcal{T}} E\left[\int_0^{\hat{\tau}} e^{-rt} \big(\delta_t - C(R_t) \big) dt \right]. \tag{4}$$

If τ^* is the optimal liquidation time, equity value at time $t < \tau^*$ is

$$W_t = E_t\left[\int_t^{\tau^*} e^{-r(s-t)} \big(\delta_s - C(R_s) \big) ds \right]. \tag{5}$$

Interpretation. Equityholders internalize the fact that a downgrade increases coupon payments. The lower the rating path, the more attractive early default becomes.

2.4 Debt value and bankruptcy costs

The market value of the ratings-based PSD obligation is

$$U_t = E_t\left[\int_t^{\tau^*} e^{-r(s-t)} C(R_s) ds \right] + E_t\left[e^{-r(\tau^*-t)} \big(A_{\tau^*} - \rho(A_{\tau^*}) \big) \right], \tag{6}$$

where $\rho(A)$ is the bankruptcy cost, increasing in asset value and smaller than the asset value itself. Upon bankruptcy, bondholders take over the firm.

Interpretation. Welfare losses come from bankruptcy costs. This becomes crucial later because all equilibria are accurate by assumption, so welfare is entirely about how often the firm is pushed into bankruptcy.

2.5 What counts as an accurate rating

The rating agency is assumed to prefer accurate ratings to inaccurate ones and to be indifferent across accurate ratings. Ratings are defined to be accurate if

$$R_t = i \qquad \text{whenever} \qquad P(\hat{\tau} - t \leq T \mid \mathcal{F}_t) \in [G_i, G_{i-1}), \tag{7}$$

for a fixed horizon T and exogenous target transition thresholds

$$G_0 = 1 > G_1 > \cdots > G_I = 0. \tag{8}$$

Higher ratings therefore correspond to lower default probabilities over horizon T.

Interpretation. Accuracy is horizon-based. The agency is not trying to infer an unobservable type; it is trying to classify default risk over a fixed future window.

2.6 Markov strategies: default threshold and rating thresholds

Because cash flow is Markov, the paper focuses on Markov Perfect Equilibria.

A **Markov default policy** takes the threshold form

$$\tau(\delta_B) = \inf\{s : \delta_s \leq \delta_B\}. \tag{9}$$

The firm defaults the first time cash flow hits the boundary δ_B.

A **Markov rating policy** is given by thresholds $H = \{H_i\}_{i=0}^{I}$ with

$$0 = H_0 < H_1 < \cdots < H_I = \infty, \tag{10}$$

so that the firm receives rating i when

$$\delta_t \in [H_{i-1}, H_i). \tag{11}$$

Interpretation. Both players choose cutoffs on the same cash-flow axis. The firm chooses where default happens; the agency chooses where each rating notch begins.

2.7 The feedback loop in one sentence

The mechanism of the paper can be summarized as:

low cash flow $\Rightarrow$ lower rating $\Rightarrow$ higher coupons $\Rightarrow$ earlier default $\Rightarrow$ higher measured default probability $\Rightarrow$ lower rating.

Interpretation. Once debt is ratings-sensitive, the rating agency cannot be treated as a passive observer. The rating policy itself changes the object that the agency is trying to measure.

3 Core model and theoretical results (all equations + interpretation)

3.1 Baseline mechanism and hypotheses

The model asks what happens when both the firm and the rating agency react optimally to each other.

- If the agency sets tougher rating thresholds, the firm faces higher coupons and wants to default earlier.
- If the firm plans to default earlier, the agency must assign lower ratings to remain accurate.
- This two-way response can produce several self-consistent fixed points.

Interpretation. The paper is not saying that agencies are biased and therefore soft. It is saying that even a purely accuracy-minded agency can support either a soft or a tough accurate equilibrium.

3.2 Firm best response: the ODE and smooth-pasting conditions

Given a rating threshold vector H, the coupon schedule becomes a step function $C^H(\delta)$, and the firm's problem is to choose the default boundary δ_B to maximize

$$\widetilde{W}(\delta_B, H) = E\left[\int_0^{\tau(\delta_B)} e^{-rt}\big(\delta_t - C^H(\delta_t)\big)dt\right]. \tag{12}$$

At each ratings region, the equity value solves the ODE

$$\frac{1}{2}\sigma^2(x)W''(x) + \mu(x)W'(x) - rW(x) + x - C^H(x) = 0. \tag{13}$$

The optimal default boundary and piecewise value-function coefficients are pinned down by:

- **value matching at default:** $W(\delta_B) = 0$,
- **smooth pasting at default:** $W'(\delta_B) = 0$,
- **continuity and smoothness at each rating threshold:** $W(H_i^-) = W(H_i^+)$ and $W'(H_i^-) = W'(H_i^+)$,
- **bounded slope of equity value.**

Interpretation. Once the rating schedule is fixed, the firm faces a standard optimal stopping problem with piecewise coupon levels. Tougher ratings work exactly like a harsher step-up debt schedule.

3.3 Rating-agency best response: accurate thresholds from default probabilities

Given a default threshold δ_B, the agency chooses rating thresholds so that each threshold cash-flow level has the target default probability over horizon T:

$$P\big(\tau(\delta_B) - t \le T \mid \delta_t = H_i\big) = G_i. \tag{14}$$

Because the first-passage probability is continuous and strictly decreasing in current cash flow, these thresholds exist and are unique.

Interpretation. Conditional on the firm's default rule, the agency's problem is simple: find the cash-flow levels at which default risk crosses the benchmark probabilities associated with each rating notch.

3.4 Equilibrium as a fixed point

Let $\delta_B(H)$ denote the firm's best-response default threshold to rating thresholds H, and let $H(\delta_B)$ denote the agency's best-response rating thresholds to default boundary δ_B. Then the set of Markov equilibria is

$$E = \left\{(x, y) \in \mathbb{R} \times \mathbb{R}^{I+1} : (x, y) = \big(\delta_B(y), H(x)\big)\right\}. \tag{15}$$

An equilibrium is therefore a pair of thresholds that are mutually best responses.

Interpretation. The central problem is not solving the firm side or the agency side separately. It is finding a fixed point of the two maps.

3.5 Strategic complementarity and existence of soft and tough equilibria

Two monotonicity results do the heavy lifting:

- **Proposition 1.** $\delta_B(H)$ is increasing in H.
- **Proposition 2.** $H(\delta_B)$ is increasing in δ_B.

A tougher rating policy raises coupon payments and makes earlier default optimal. Earlier default raises true default probabilities and forces the agency to use tougher thresholds to stay accurate.

These increasing best responses imply strategic complementarity. By Tarski's fixed-point logic, the game has a largest and a smallest Markov equilibrium:

- **soft-rating-agency equilibrium:** the smallest fixed point, with low δ_B and low rating thresholds;
- **tough-rating-agency equilibrium:** the largest fixed point, with high δ_B and high rating thresholds.

There can also be intermediate equilibria.

Interpretation. The model is structurally similar to self-fulfilling crisis models. Expectations embodied in rating policy can become self-confirming because they feed into the firm's own default choice.

3.6 Algorithm 2, convergence, and a uniqueness test

The paper computes equilibria by iterated best response. Starting from an arbitrary default boundary x_0:

1. compute $H(x_{n-1})$ from the accuracy condition above;
2. compute $x_n = \delta_B\big(H(x_{n-1})\big)$ from the firm's optimal stopping problem;
3. iterate until convergence.

The paper proves:

- the algorithm always converges to an equilibrium;
- starting from the lowest boundary yields the soft equilibrium;
- starting from the highest boundary yields the tough equilibrium.

Therefore, the game has a unique Markov equilibrium if and only if the two starting points converge to the same fixed point.

Interpretation. The algorithm is both a computational tool and a characterization theorem. It tells you not only that equilibria exist, but how to locate the extreme equilibria and how to test uniqueness.

3.7 Benchmark with no feedback: fixed-coupon debt

If the debt contract is a fixed-coupon consol bond, so $C(i) = c$ for all ratings, then ratings do not affect financing costs. In that case, the equilibrium is unique.

Interpretation. This benchmark isolates the feedback channel. Without ratings-sensitive coupon payments, the agency is just estimating the first-passage distribution through a constant default boundary, and the multiplicity problem disappears.

3.8 Welfare ranking and equilibrium selection

The model assumes ratings are accurate in every equilibrium. Therefore, welfare differences do *not* come from informational distortions; they come only from bankruptcy costs.

A higher default boundary means the firm defaults earlier and more often, which raises expected bankruptcy costs. Hence:

- the **tough equilibrium** is the worst equilibrium,
- the **soft equilibrium** is the best equilibrium.

The policy discussion follows directly:

- **Issuer-pay model.** If the agency receives a small flow fee while the borrower survives, then among accurate rating policies it prefers the one with the lowest default probability. That selects the soft equilibrium.
- **Stress tests.** If the agency asks whether the firm would survive a severe scenario in which rating triggers fire, it may effectively evaluate the firm under the tough equilibrium and select exactly the equilibrium that destroys value.

Interpretation. In this framework, "soft" does not mean "inflated." It means choosing the self-consistent accurate equilibrium with the fewest bankruptcies.

3.9 Stability and the credit-cliff dynamic

If the equilibrium is unique, the paper shows that it is globally stable under best-response dynamics. Starting from any Markov strategy, iterative adjustment converges to the unique equilibrium, so small shocks generate gradual rating changes.

When multiple equilibria exist, however, some equilibria can be unstable. Then a small shift in fundamentals or a small deviation in one player's response can set off a feedback loop:

slightly tougher rating policy $\Rightarrow$ earlier default $\Rightarrow$ lower accurate rating $\Rightarrow$ even earlier default.

This can produce:

- multi-notch downgrades,
- jumps from a high rating to immediate default,
- and in extreme cases a "death spiral."

Interpretation. Sudden rating collapses do not necessarily require irrationality or agency slack. They can arise endogenously from equilibrium selection in a perfectly rational model.

3.10 Competition between rating agencies

The paper next studies two rating agencies, indexed by $k \in \{1, 2\}$, with ratings R_t^k. The coupon schedule becomes $C(R_t^1, R_t^2)$ and is decreasing in both ratings.

Each agency still wants accuracy, but it also values being *more accurate than the rival*. This captures competition for market share or reputation.

The first result is symmetry: in equilibrium, both agencies choose the same threshold vector, and those common thresholds must lie in the single-agency equilibrium set E.

But equilibrium selection now depends on how the contract resolves split ratings:

- if coupon payments depend on the **minimum** of the two ratings, the unique Markov equilibrium is the **tough** equilibrium;
- if coupon payments depend on the **maximum** of the two ratings, the unique Markov equilibrium is the **soft** equilibrium.

Interpretation. Under a minimum-rating contract, one agency can unilaterally make funding more expensive by rating more harshly. That tougher deviation raises default risk and can make the rival inaccurate, so the logic of competition pushes the system toward the tough equilibrium. Competition therefore worsens outcomes precisely when contract design gives the harsher agency unilateral power.

3.11 Comparative statics on the set of equilibria

For both the soft and the tough equilibria, the paper shows that the default boundary δ_B and the rating thresholds H are:

- increasing in coupon payments C,
- increasing in the interest rate r,
- decreasing in the drift $\mu(\cdot)$ of cash flows,
- decreasing in the target rating thresholds G.

The intuition is straightforward:

- higher coupons or a higher discount rate make continuation less attractive, so the firm's best response shifts toward earlier default;
- stronger fundamentals shift the system toward later default and softer ratings;
- looser target default-probability cutoffs allow the agency to maintain higher ratings at any given default boundary.

Important caveat. These are comparative statics *on the equilibrium set.* A small parameter change can also eliminate equilibria altogether, leaving only the tough equilibrium. So a small shock can generate a large discrete outcome.

3.12 Numerical computation and quantitative examples

The paper solves the model numerically in two cases.

1. Geometric Brownian motion. If cash flow follows

$$d\delta_t = \mu\delta_t\, dt + \sigma\delta_t\, dB_t, \tag{16}$$

then the first-passage-time distribution is inverse Gaussian, the rating-threshold map is linear in the default boundary, and the equilibrium is unique.

Interpretation. With proportional drift and volatility, the feedback effect remains, but it is not strong enough in the paper's setup to generate multiplicity.

2. Mean-reverting cash flow. If cash flow follows

$$d\delta_t = \lambda(\bar{\mu} - \delta_t)dt + \sigma\delta_t\, dB_t, \tag{17}$$

then the model can generate multiple equilibria. The first-passage distribution is computed by Monte Carlo simulation.

In the paper's base mean-reverting example,

$$r = 0.06, \qquad \lambda = 0.15, \qquad \bar{\mu} = 1, \qquad \sigma = 0.4,$$

with two ratings, coupon levels $c_1 = 1.3$ and $c_2 = 0.75$, and target threshold $G_1 = 20\%$, there are three equilibria. The present value of bankruptcy costs is approximately:

- **close to zero** in the soft equilibrium,
- about **10% of firm asset value** in the tough equilibrium when bankruptcy destroys 20% of

assets.

Interpretation. The welfare gap is not just a theoretical curiosity. Under plausible mean-reverting dynamics, equilibrium selection has quantitatively large consequences.

4 Equilibrium logic and policy interpretation (what makes the results credible)

4.1 Why both soft and tough ratings can be accurate

At first sight it may sound contradictory that two different rating policies can both be accurate. The resolution is that the object being measured — default probability — is endogenous to the rating policy itself.

A tougher rating increases coupons, which makes default more likely. Once the firm responds that way, the tougher rating becomes accurate. A softer rating lowers coupons, keeps the firm alive longer, and is also accurate. Accuracy is therefore an equilibrium concept, not a primitive property of the firm.

Interpretation. The paper's main insight is not about agency bias. It is about the endogenous nature of credit quality under ratings-sensitive financing.

4.2 Why issuer-pay can improve welfare in this model

The standard critique of issuer-pay is straightforward: if the firm pays, the agency may inflate ratings. Manso's argument is subtler. If the fee is *small* relative to reputational concerns, the agency still primarily wants accuracy. The fee only serves as a tie-breaker across accurate equilibria by favoring borrower survival.

Interpretation. The model does *not* claim that issuer-pay is always harmless. It claims that a very small survival-contingent issuer fee can improve equilibrium selection when the underlying problem is multiplicity among accurate rating policies.

4.3 Why stress tests can backfire

Stress tests seem naturally conservative, but in this model they can be too conservative in exactly the wrong way. If the test asks whether the firm can survive after triggers fire, it implicitly evaluates the firm under a tougher rating regime. That can push the agency toward the equilibrium in which the firm indeed fails.

Interpretation. The model separates two ideas that are often conflated:

- incorporating rating-trigger exposure into ratings is necessary for accuracy;
- evaluating the firm under the harshest self-fulfilling scenario is *not* necessarily optimal.

4.4 Why competition can worsen outcomes instead of improving them

In many settings, competition disciplines intermediaries. Here the result depends on contract form. Under minimum-rating contracts, a single agency can unilaterally make financing terms worse by rating more harshly. That harsher action changes actual default risk and can make the rival's softer rating inaccurate.

Interpretation. Competition is not bad per se. The problem is competition combined with a contract rule that gives the lowest rating immediate contractual force.

4.5 What assumptions are doing the work

Several assumptions are essential:

- **ratings-based PSD is crucial;** without it, there is no feedback loop;
- **default is endogenous;** otherwise ratings would not affect the timing of bankruptcy;
- **Markov strategies matter,** especially for the competition result, because richer repeated-game strategies could potentially sustain other outcomes;
- **bankruptcy costs are the only inefficiency,** which is why the soft equilibrium is always best in the baseline model.

Interpretation. The paper is intentionally focused. It isolates the feedback mechanism cleanly, but that also means some real-world frictions are held fixed.

4.6 What the paper does not model

The paper leaves several issues outside the baseline setup:

- it takes the existence of ratings-sensitive debt as given rather than deriving the firm's capital-structure choice inside the model;
- it does not model the usual investor-information conflict or structured-finance business-line incentives directly;
- it does not study systemic spillovers across firms, only the feedback loop for one borrower;
- it notes that if other frictions matter — for example, if default helps replace negligent managers — the tough equilibrium need not always be welfare-dominated.

Interpretation. The paper is a mechanism paper, not a full institutional model of the rating industry.

4.7 Relation to the literature and what is new

The paper sits at the intersection of three literatures:

- endogenous-default structural credit-risk models;
- performance-sensitive debt models;
- self-fulfilling crisis models with strategic complementarities.

Its main novelty is to combine ratings-based debt with an accuracy-motivated rating agency and endogenous default, thereby showing how multiple accurate equilibria and policy distortions can arise even without the classic rating-inflation agency problem.

5 Tables and figures

5.1 Figure 1: multiple equilibria from crossing best responses

Read.

- In the two-rating example, the firm's best-response default boundary and the agency's best-response rating threshold can intersect three times.
- The left intersection is the soft equilibrium, the right intersection is the tough equilibrium, and the middle intersection is an unstable intermediate equilibrium.

Economic message. The central mechanism of the paper is visible in one picture: mutually reinforcing best responses can create both good and bad self-fulfilling equilibria.

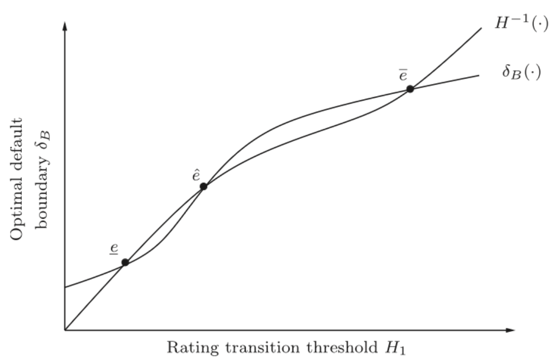

Fig. 1. The figure plots best-response functions of the rating agency and the borrowing firm for the case in which there are two possible credit ratings (I=2). Points $\underline{e}$, $\hat{e}$, and $\bar{e}$ are Markov equilibria of the game. The soft-rating-agency equilibrium is given by $\underline{e}$, while the tough-rating-agency equilibrium is given by $\bar{e}$. The point $\hat{e}$ corresponds to an intermediate equilibrium.

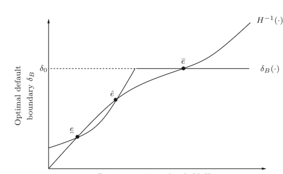

Fig. 2. The figure plots best-response functions of the rating agency and the borrowing firm and the corresponding equilibria for the case in which there are two possible credit ratings (I=2). In the situation illustrated in the figure, the borrowing firm would fail a stress test, since the tough-rating-agency equilibrium $\bar{e}$ involves immediate default. The firm would survive if the rating agency selected the soft-rating-equilibrium.

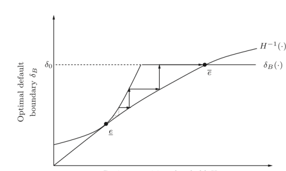

Fig. 3. The figure plots best-response functions of the rating agency and the borrowing firm and the corresponding equilibria for the case in which there are two possible credit ratings (I=2). The soft-rating-agency equilibrium $\underline{e}$ is unstable. Small shocks may produce a "credit-cliff dynamic" that leads to the tough-rating-agency equilibrium $\bar{e}$, which in this case involves immediate default.

5.2 Figure 2: stress tests can select the wrong equilibrium

Read.

- The figure shows a case in which the tough equilibrium implies immediate default, so the borrower would fail a stress test based on triggered covenants.

- Yet the soft equilibrium is also accurate and allows the borrower to survive.

Economic message. A stress test can push the agency toward the worst equilibrium precisely when a welfare-superior accurate equilibrium exists.

5.3 Figure 3: unstable soft equilibrium and the credit-cliff dynamic

Read.

- The soft equilibrium is locally unstable.
- A small increase in rating toughness triggers iterative responses that march the system toward the tough equilibrium, which in the picture involves immediate default.

Economic message. Large discrete downgrades can emerge from very small shocks once the economy sits near an unstable equilibrium.

5.4 Figure 4: a small shock can eliminate the good equilibria

Read.

- Both the soft and tough equilibria may initially be locally stable.
- A small shock, such as a higher interest rate, can wipe out the soft and intermediate equilibria so that only the tough equilibrium remains.

Economic message. Credit cliffs need not require knife-edge local instability; they can arise because a small parameter shift changes the equilibrium set discontinuously.

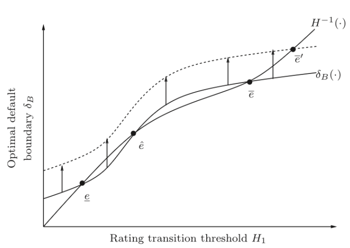

Fig. 4. The figure plots best-response functions of the rating agency and the borrowing firm and the corresponding equilibria for the case in which there are two possible ratings (I=2). A small shock to fundamentals may eliminate all equilibria except for the tough-rating-agency equilibrium $\bar{e}'$, leading to a multi-notch downgrade or even immediate default.

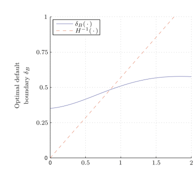

Fig. 5. The figure plots best-response functions of the rating agency and the borrowing firm and the corresponding equilibrium for the case in which the cash-flow process follows a geometric Brownian motion. The parameters used to plot the figure are r=0.06, $\mu = 0.02$, $\sigma = 0.25$, I=2, $c_1 = 1$, c_2=1.5, and $G_1 = 2\%$.

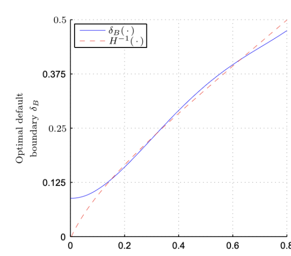

Fig. 6. The figure plots best-response functions of the rating agency and the borrowing firm and the corresponding equilibrium for the case in which the cash-flow process follows the mean-reverting process (11). The parameters used to plot the figure are r=0.06, $\lambda = 0.15$, $\mu = 1$, $\sigma = 0.4$, I=2, c_1=1.3, c_2=0.75, and $G_1 = 20\%$.

5.5 Figure 5: geometric Brownian cash flows yield a unique equilibrium

Read.

- Under geometric Brownian cash flows, the best-response curves intersect once.
- The figure is the paper's clean benchmark without multiplicity.

Economic message. Feedback effects do not mechanically imply multiple equilibria. The shape of the cash-flow process matters.

5.6 Figure 6: mean reversion generates large welfare differences

Read.

- With mean-reverting cash flows, the best-response curves can intersect three times.
- The soft equilibrium has almost no expected bankruptcy losses, while the tough equilibrium has bankruptcy losses worth roughly 10% of asset value in the paper's calibration.

Economic message. The model's multiplicity is quantitatively important, not just a conceptual possibility.

6 Short summary

- Credit ratings can be self-fulfilling because they change financing costs and therefore change actual default risk.
- The firm and the rating agency form a game of strategic complementarity, which generates a soft equilibrium and a tough equilibrium.
- Both equilibria can be accurate, but the soft equilibrium is welfare-superior because it minimizes bankruptcy costs.
- Small issuer fees can help the agency select the soft equilibrium; stress tests can unintentionally select the tough one.
- With multiple equilibria, small shocks can create a credit-cliff dynamic with multi-notch downgrades or immediate default.
- Competition between agencies worsens outcomes under minimum-rating contracts and improves outcomes under maximum-rating contracts.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

Credit ratings are not merely signals about default risk. When debt contracts depend on ratings, ratings themselves shape default risk. That feedback can generate multiple accurate equilibria, large welfare differences across equilibria, and nontrivial policy trade-offs in the rating industry.

7.2 Economic intuition

The key intuition is simple: a downgrade can make itself true. By raising coupon payments or worsening refinancing terms, a lower rating reduces equityholders' incentive to continue, which raises default probability and validates the downgrade. A softer rating can be equally accurate if it supports a lower-cost path that keeps the borrower alive.

7.3 Takeaway

The paper turns a common criticism of rating agencies on its head. In a world with feedback effects, being harsher is not always more accurate and not always better. The central policy problem is how to select the good accurate equilibrium rather than how to push agencies to be maximally tough.